Lab 07 - Build a DORA and Problem-Management Evidence Dashboard

Course: DevOps Foundation Training (TGS-2024049184) Version: v4.0 Topic: 07 Estimated time: 35-45 minutes

Workplace scenario

Leadership wants one slide of delivery-performance evidence, while engineering needs definitions precise enough to reproduce the numbers.

Objective

Apply K5 and A6 by calculating documented delivery metrics and linking them to tracked problem-management actions.

What you will produce

A dora_metrics.csv, problem_actions.csv and dora_dashboard.png with metric definitions.

Prerequisites

- R 4.x available as Rscript.
- Work from this lab folder so relative paths resolve.
- Use only the supplied synthetic data; no internet connection or external package is required.
- Preserve the source data. Make experimental changes in a copied file, then restore it.

Folder map

```
Lab-07-dora-problem-dashboard/
■■■ README.md / README.pdf
■■■ CHECKLIST.md / CHECKLIST.pdf
■■■ dora_problem_dashboard.R
■■■ data/
■■■ output/            # reproducible evidence produced by the R script
```

Detailed procedure

Step 1 - Review deployments.csv and problem_actions.csv; confirm each metric has the fields needed for reproducible calculation.

```
Rscript -e 'str(read.csv("data/deployments.csv")); str(read.csv("data/problem_actions.csv"))'
```

Record what you observed before continuing.

Step 2 - Run dora_metrics.R to calculate throughput, instability and action-tracking measures.

```
Rscript dora_metrics.R
```

Record what you observed before continuing.

Step 3 - Inspect dora_metrics.csv and compare the current five-metric DORA model with the historical four-key vocabulary in the v3.4 guide.

```
Rscript -e 'print(read.csv("output/dora_metrics.csv"))'
```

Record what you observed before continuing.

Step 4 - Open dora_dashboard.png and identify one improvement hypothesis; do not rank individuals or unrelated services.

```
open output/dora_dashboard.png
```

Record what you observed before continuing.

Step 5 - Review problem_actions.csv for overdue or unverified corrective actions.

```
Rscript -e 'print(read.csv("output/problem_actions.csv"))'
```

Record what you observed before continuing.

Step 6 - Record the metric definition, data source, owner and next review date in CHECKLIST.md.

```
open CHECKLIST.md
```

Record what you observed before continuing.

Evidence to retain

[ ] output/dora_metrics.csv

[ ] output/problem_actions.csv

[ ] output/dora_dashboard.png

[ ] completed CHECKLIST.md

Acceptance criteria

- Five current DORA metrics
- Definitions and denominators explicit
- Open actions tracked
- Improvement hypothesis avoids individual ranking

Verification

Five DORA measures are present; the deployment counts reconcile; every open action has an owner and due date; dashboard values match the CSV.

Troubleshooting

- cannot open file: confirm your terminal is in this lab folder.
- Rscript: command not found: install R, reopen the terminal and run Rscript --version.

- A validation stop(...) message is evidence that the input broke a stated rule; inspect the named field rather than bypassing the check.
- If an output looks wrong, compare row counts and IDs with the source CSV before changing the analysis.

Clean rerun

Delete only the files inside output/, then run the R script again. The source CSV and script remain unchanged.